COMETFWC067
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Aim

To interface the ESP32-WROOM development board with a 7447 BCD-to-7-Segment Decoder IC and a common anode 7-segment display by implementing the simplified Boolean expression through the ESP32 GPIO pins, and to verify on hardware that the 7-segment display shows the correct digit for each input combination, in agreement with the derived truth table.

GATE Question

The Boolean expression for the output f of a 4:1 multiplexer with inputs I0 = R, I1 = R', I2 = R', I3 = R and select lines S1 = P, S0 = Q is one of the following:

Option	Expression
A	(P xor Q xor R)'
B	P xor Q xor R
C	P + Q + R
D	(P + Q + R)'

Question Analysis

1. Select-Line to Output Mapping

S1 (P)	S0 (Q)	Line Selected	Output f
0	0	I0	R
0	1	I1	R'
1	0	I2	R'
1	1	I3	R

2. Sum of Products Form

f = P'Q'R + P'QR' + PQ'R' + PQR

3. Observation and Final Expression

f is 1 only where P, Q, R contain an odd number of 1's (001, 010, 100, 111) - the defining property of the three-input XOR operation.

Therefore, f = P xor Q xor R, which matches Option (B).

Correct Answer: (B) f = P xor Q xor R

Components Used

S.No.	Component
1	ESP32-WROOM Development Board
2	USB Cable
3	Breadboard
4	7447 IC
5	Common Anode 7-Segment Display
6	Jumper Wires

Purpose of Each Component

Component	Function
ESP32-WROOM Development Board	Runs the flashed firmware and generates the GPIO output corresponding to the simplified Boolean expression.
USB Cable	Connects the ESP32 board to the computer for flashing the firmware and powering the board.
Breadboard	Provides a platform to build and hold the circuit connections.
7447 IC	Decodes the BCD input from the ESP32 GPIO pins into the signals needed to drive the 7-segment display.
Common Anode 7-Segment Display	Displays the decimal digit corresponding to the decoded output of the 7447 IC.
Jumper Wires	Connects the ESP32 GPIO pins to the 7447 IC and the display circuit on the breadboard.

Hardware Connections

ESP32-WROOM to 7447 IC Connections

S.No.	ESP32-WROOM Pin	7447 IC Pin
1	GPIO16	A
2	GPIO17	B
3	GPIO18	C
4	GPIO19	D
5	VIN (5V)	VCC (Pin 16)
6	GND	GND (Pin 8)

7447 IC to Common Anode 7-Segment Display Connections

S.No.	7447 Output	7-Segment Display
1	a	Segment a
2	b	Segment b
3	c	Segment c
4	d	Segment d
5	e	Segment e
6	f	Segment f
7	g	Segment g
8	Common Anode	+5 V
9	Segment Pins	Through 220 Ω Resistors

Programming Connections

S.No.	Component	Connection
1	ESP32 USB Port	Ubuntu PC USB Port
2	USB Cable	Power and Program Upload
3	Ubuntu PC	PlatformIO and ESPTool

Execution Procedure

1. A PlatformIO project was set up for the ESP32-WROOM board (board type esp32dev).

2. The source code implementing the Boolean expression f = P xor Q xor R on GPIO 2 was placed inside the src folder of the project.

3. The project was built to generate the firmware, bootloader, and partition binary files (bootloader.bin, partitions.bin, firmware.bin).

4. The ESP32 board was connected to the Ubuntu PC through a USB cable, and the system was checked to confirm the board was detected as a serial device (/dev/ttyUSB0).

5. The chip identity was verified using the esptool.py chip_id command.

6. The generated binary files were flashed onto the ESP32 at their respective memory offsets using esptool.py write_flash.

7. Once flashing completed, the EN (reset) button on the board was pressed to run the new firmware.

8. The output was observed to confirm that its behaviour matched the expected logic output of the Boolean expression.

Result

The simplified Boolean expression f = P xor Q xor R was successfully implemented on the ESP32-WROOM development board using the 7447 IC and a common anode 7-segment display. The GPIO outputs of the ESP32 were fed to the 7447 IC, which decoded them and drove the display accordingly. The displayed output was observed for all input combinations of P, Q, and R and was found to be in agreement with the truth table derived from the GATE question, thereby validating the correctness of the implemented Boolean function.